

Azmi Rehman Khan

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SUMMARY

Autonomous Vehicle Engineering student specializing in **Computer Vision** , **AI-driven systems** , and **full-stack data architecture** . Experienced in developing custom **machine learning pipelines** , low-latency **streaming architectures** , and interactive data interfaces using **Python** , **FastAPI** , and **Streamlit** . Highly focused on transitioning complex engineering logic into intelligent, decoupled software infrastructure for pure tech and AI applications.

EDUCATION

B.ENG in Autonomous vehicle engineering

Technische Hochschule Ingolstadt

09/2024 – Present

Ingolstadt, Germany

Core Modules: Algorithms & Data Structures, Foundations of Machine Learning, Digital Signal Processing, Sensor Data Processing & Fusion, Model-based SW Engineering.

PROJECTS

Race Strategy AI: RAG Telemetry Dashboard 

Python, FastAPI, Streamlit, LangChain, Gemini API, Prompt Engineering

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to ingest and parse the FIA Formula 1 Sporting Regulations, reducing manual rule-verification time by **80%** .
- Built a **decoupled dual-engine architecture** featuring a localized **REST API backend (FastAPI)** and an interactive web interface (**Streamlit**) handling **50+** simultaneous API requests.
- Integrated the **Google Gemini API** via **LangChain** , utilizing strict persona prompting and persistent local file memory to ensure **95% context-aware query accuracy** .

Esports EV Hunter - Autonomous AI Betting Analytics Pipeline 

Python, FastAPI, Docker, n8n, LangGraph, REST APIs

- Engineered a **containerized microservices architecture** using **Docker** and **FastAPI** to autonomously ingest and process over **1,000 real-time JSON data points** from live sports APIs.
- Developed an **AI agent state machine** utilizing **LangGraph** to evaluate mathematical probability models, successfully identifying **Expected Value (EV) discrepancies** across **multiple bookmakers** with **zero manual intervention**.
- Orchestrated a **24/7 automated data workflow** via **n8n** to parse complex analytics and dispatch dynamically generated **HTML email reports** via **SMTP** , eliminating an estimated **10+ hours** of manual market analysis per week.

GestureVision — Real-Time Gesture Controlled Vision System 

Python, OpenCV, MediaPipe, FastAPI, Streamlit, WebSockets

- Engineered a **real-time computer vision pipeline** for gesture-controlled ROI filtering using **MediaPipe** hand tracking and **OpenCV** , processing video feeds at **30 frames per second** .
- Architected a low-latency **WebSocket-based FastAPI backend** to stream and process live webcam frames dynamically with **sub-50ms latency** .
- Developed a cyberpunk-inspired **Streamlit HUD interface** with real-time gesture interaction and live visual filter switching.
- Implemented dynamic **Region-of-Interest (ROI) extraction** , pinch gesture recognition, and multiple **image processing filters** including thermal, sketch, grayscale, and inversion effects.

F1 Race Analyzer — Formula 1 Data Visualization Web App

Live Telemetry Processing & Interactive Dashboarding

- Architected an **interactive Streamlit application** utilizing the **FastF1 library** and **Plotly** to visualize over **100,000 rows of live telemetry** and lap-time progression data per session.
- Engineered robust **data handling capabilities** , optimizing caching mechanisms to reduce application load times by **40%** .
- Automated **chart generation and export functionality** to streamline presentation and performance analysis for large-scale telemetry datasets.

SKILLS		
Languages & Frameworks Python, C++, FastAPI, Streamlit, React, HTML.	Data Architecture & AI RAG Pipelines, LangChain, LLM Applications, ETL Pipelines, Vector Embeddings, Prompt Engineering, LangGraph, AI Agents, Automated Workflows	Technical / DSP Digital Filtering, Frequency Response Analysis, Spectral Signal Processing (DFT/IDFT), Fourier Series & Nyquist Sampling, Circuit Design
Data & Analysis Data Visualization, Time/Frequency-Domain Plotting, Multi-Filter Signal Strategies, Plotly, Pandas, NumPy, API Ingestion, JSON Parsing, Statistical Modeling (EV)	Tools & Infrastructure Docker, n8n, Git, WebSockets, REST APIs.	

PROFESSIONAL EXPERIENCE	
Rider <i>Flink</i> Developed strong reliability, time management, and adaptability while performing high-volume deliveries using digital logistics systems in a fast paced environment.	06/2024 – Present Nurnberg, Germany

LANGUAGES	
English C1	German A2